

Intelligent Intrusion Detection System (IIDS) using Hybrid Machine Learning

Abstract-Due to the rapid development of interconnected networks, cybersecurity risks have significantly evolved, and intrusion detection systems (IDS) have become necessary for modern cybersecurity systems. The traditional methods of IDS have become limited in their ability to detect complex cyber-attacks. In response, we have proposed an intelligent intrusion detection system (IIDS) that utilizes a hybrid machine learning system combining Random Forest (RF) and a deep learning structure of Convolutional Neural Networks (CNNs) and Long Short-Term Memory (LSTM) networks. We argue that Random Forest is one of the best methods for feature selection and classification and for the detection of network traffic patterns, the CNN-LSTM structure is one of the best methods due to its ability to detect spatial and temporal relationships. The NSL-KDD dataset was chosen to facilitate training and testing for the model, as it maintains a well-balanced representation of normal and anomalous network traffic. We utilize various preprocessing methods, including normalization, encoding, and feature extraction, in order to improve the efficacy of the model. Through initial testing, we believe that hybrid machine learning has significantly improved detection efficacy and generalization, in addition to reducing false positives. Furthermore, we believe that the combination of machine learning and deep learning is the reason that we were able to detect both known and unknown attack patterns with a higher degree of efficacy than previous methods. In summary, we have attempted to illustrate the need for adaptive intelligent machine learning intrusion detection systems in response to the evolving cyber threat landscape.

I. INTRODUCTION

The development of digital technology has created a larger surface for cyber threats. These threats increase daily, which makes the task of protecting a digital environment very complex. An Intrusion Detection System (IDS) seeks to protect a digital environment by detecting and responding to all forms of network breaches [1]. The most common systems have either a signature-based or anomaly-based approach [9]. These older systems do not have the flexibility to counter new and complex threats, which causes them to generate a large number of false positives [1], [3].

New systems previously described have either not adopted new techniques or cannot stop complex and evolving multi-layer attacks and techniques [4]. The development of Artificial Intelligence (AI) and Machine Learning (ML) has presented new opportunities in the development of IDS systems. ML techniques such as Random Forests (RF) are designed to improve classification and predictive accuracy [1], [9]. DL systems such as Convolutional Neural Networks (CNN) and Long Short-Term Memory Networks (LSTM) are capable of capturing the deeper and rich relationships and patterns of time series data [5], [7].

This paper describes the design of the hybrid intelligent IDS (IIDS) that combines RF and CNN-LSTM [5], [6]. The RF model is used to classify and select more relevant features, while the LSTM model is used to study and find patterns in network traffic. The NSL-KDD dataset is used to validate the proposed system as it has been improved and widely used for training and testing [2], [8].

This study aims to design a system with greater detection capability, fewer false alarms, and flexible to ever-changing cyber threats [3]. The proposed system incorporates a mix of conventional machine learning and deep learning due to the fact that such an approach is

likely to improve performance and the reliability of intrusion detection systems.

II. LITERATURE REVIEW

A. Evolution of Intrusion Detection Systems

The rapid development of cloud services, Internet of Things (IoT) architectures, and distributed networks has exponentially increased the volume and complexity of network traffic [3]. As a result, signature-based Intrusion Detection Systems (IDS) that are generally based on static databases of recognized threat vectors become increasingly poor performing [9]. However, these legacy systems are extremely vulnerable to zero-day attacks and complex multi-step intrusion tactics [4], [3]. The current literature emphasises the need to move from reactive, rule-based systems to dynamic, adaptive systems using ML and DL paradigms.

B. Dataset Benchmarking and NSL-KDD

Selecting a quality training corpus is a fundamental building block for building a strong ML-based intrusion detection solution. The NSL-KDD dataset continues to be a good benchmark of various types of IDS models in the industry [2], [8]. Research evaluating heterogeneous algorithms against network traffic anomalies always confirms that the NSL-KDD dataset is an efficient method to train models learning advanced attack classifications, i.e., Denial of Service (DoS), Probe, User-to-Root (U2R) and Remote-to-Local (R2L) threats [9]. With 41 such features, the researchers found that the NSL-KDD dataset enables fine-grained training of very accurate models and reduces redundancy in records compared to past iterations such as KDD Cup '99 [2], [9].

C. Resolving Latency with Shallow Learning

Although deep learning offers detection accuracy, in real time network monitoring, the central limitation is the computational latency. Classic deep learning techniques have high computational overhead, hence are not suitable for high-throughput, early-stage traffic filtering. In order to deal with these issues, recent research supports the incorporation of Shallow

Learning algorithms, notably the Random Forest (RF), as the front-line classification mechanism [1], [9]. Powerful Random Forest methodology can tackle latency bottlenecks by providing near-instantaneous feature detection and signature matching [9]. With RF, applications can quickly segment known high-throughput threats at ultra-light latency, sparing computationally expensive deep learning resources for ambiguous or anomalous traffic.

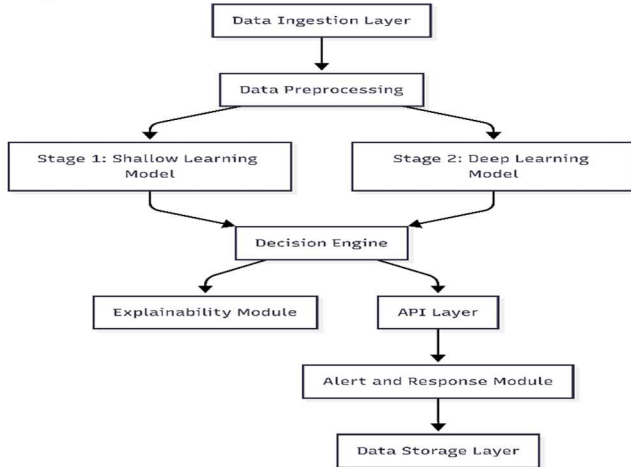
D. Hybrid Architectures for Zero-Day Mitigation

To address zero-day threats and nonlinear, multi-level attack patterns, contemporary architectures are employing multi-tiered, multi-layer, hybrid algorithms. These hybrid systems build on one another for the most processing power per unit detection time. Recent developments reveal certain benefits of the Convolutional Neural Network and Long Short-Term Memory (CNN-LSTM) hybrid architectures in network security [5], [6], [7]. The CNN function performs well taking advantage of local spatial features and co-occurring signatures in the packet headers and payloads [5], [7]. At the same time, it incorporates temporal and sequentially associated network sequence dependencies like the LSTM component [5], [6]. As contemporary cyber threats tend to be hierarchical with long time spans, from slow-rate DoS attacks to stealthy reconnaissance probing, the temporal awareness afforded by LSTM algorithms can drastically reduce false positive rates.

III. PROPOSED SYSTEM ARCHITECTURE

The backend design for the Hybrid Intrusion Detection System (IDS) is conceptualized as a multi-stage intelligent processing pipeline that merges shallow and deep learning techniques. Shallow techniques include a handful of traditional models, models with a limited number of hidden layers like Random Forest, and SVM, coupled with some threat detection [1], [9]. In contrast, deep learning techniques include the use of LSTM and CNN [5], [6]. This

combination is aimed at achieving superior detection accuracy, reduced latency, and a high degree of adaptability to evolving cyber threats. The use of Python and the Flask framework in this implementation places a high premium on modularity, scalability, and real-time responsiveness.



A. Data Ingestion and Preprocessing Layer

The Data Ingestion Layer is where the system begins, which entails the collection and preprocessing of raw network traffic and batch datasets. The system can utilize libraries like Scapy or PyShark to stream packets live or analyse previously captured packet (PCAP) files. Data sources include packet sniffers, logs, or datasets obtained externally. The ingestion process is optimized to accommodate high-velocity data streams while introducing the least latency during data collection.

Preprocessing begins with a set of transformations for model representation. These include data cleansing, normalization, and feature extraction, which result in the removal of extraneous and duplicate data, the scaling of feature values, and the extraction of relevant attributes such as packet size, protocol type, flow duration, and statistical patterns in traffic [8]. Quality control during the preprocessing stage determines the effectiveness of the model, as both machine and deep learning performance are reliant on the quality of the input data, as well as the representation of that data [2], [9].

B. Stage 1: Shallow Learning Model (Random Forest)

The initial analysis stage utilizes a Random Forest classifier, which is both clear and efficient for this function [1], [9]. This stage is intended to learn and classify attack patterns via signature learning and networks of statistics. The main goal of this stage is to function as a fast mechanism for identifying and classifying well-known and common intrusions, such as denial-of-service attacks, port scanning, and brute-force attempts [1]. The system efficiently addresses these common occurrences early in the pipeline, minimizing the workload on subsequent deep learning stages.

Random Forest is effective for rapid intrusion detection and in real-time systems [9]. It is effective in high dimensional data with limited overfitting and fast inference time [9]. Data that is clearly classified in this stage is sent to the Decision Engine, while data that is uncertain or unclear is analysed further in the subsequent stages.

C. Stage 2: Deep Learning Model (CNN-LSTM)

The second stage employs a hybrid deep learning model that merges Convolutional Neural Networks (CNN) with Long Short-Term Memory (LSTM) networks [5], [6], [7]. This model is custom-built to capture both spatial and temporal aspects of network traffic data. The CNN part captures spatial features by finding local patterns and correlations within the input feature, such as the inter-relationships of fields in a protocol or attributes of packets [5], [7]. These features are then passed to the LSTM model, which analyses the patterns of data over time and identifies patterns in sequences that assist in detecting complex and novel types of attacks [5].

This phase is particularly beneficial to identify novel and atypical attacks that do not conform to known patterns. While it does demand greater computational resources than the first stage, it is not used frequently, enabling the system to operate efficiently without compromising on precision [6].

D. Explainability Module (XAI – SHAP/LIME)

To enhance system usability and reliability, an Explainability Module has been integrated [3]. This module employs SHAP, LIME, and other techniques to provide an explanation of a model prediction and the contribution of specific features to that prediction. For each detected attack or some deviant behaviour, system analysts will be able to follow the rationale behind the system assertion. This becomes critical in sensitive systems, where accountability and auditability of a system decision become of paramount importance [3].

E. Decision Engine

The Decision Engine integrates input from all phases and assigns the final verdict as to the case. For Random Forest and CNN-LSTM, the confidence score and the threshold cutoffs are considered [9]. Model concurrence, prediction probability, and uncertainty computation determine the final classification. Where model outputs are consistent, predictions are clear. Conversely, where model outputs are inconsistent, the system operates to the benefit of the deep learning model in particular. This is notably the case in nuanced, difficult predictions, fostering confidence in the system and reducing the frequency of incorrect predictions.

IV. METHODOLOGY AND DATASETS

A. Dataset Description

This study employs the NSL-KDD dataset, a refined version of the original KDD Cup 1999 benchmark, widely regarded as the standard evaluation corpus for network intrusion detection research [2], [8]. The dataset comprises 125,973 training records and 22,544 test records, each described by 41 input features and a class label [2]. The 41 features are grouped into three functional categories: basic features derived from individual TCP/IP connections (e.g., duration, protocol type, service, flag, source and destination bytes), content-based features extracted from the data payloads of connections (e.g., number of failed logins, root shell access attempts), and traffic-based features computed using a two-second

time window (e.g., same-host and same-service connection counts, error rates) [2], [9]. The class label identifies each record as either normal traffic or one of four attack categories: Denial of Service (DoS), Probe, Remote-to-Local (R2L), and User-to-Root (U2R) [2], [9]. A notable characteristic of NSL-KDD is its severe class imbalance: DoS and Probe attacks are well-represented with tens of thousands of instances, whereas R2L and U2R categories collectively account for fewer than 2,000 training records, presenting a significant challenge for multi-class classifier training [8].

B. Preprocessing Pipeline

Raw network traffic data requires systematic preprocessing before it can be reliably used to train machine learning models [8]. The preprocessing pipeline adopted in this work consists of four sequential stages.

Missing Value Handling: Although NSL-KDD contains no explicitly missing values, zero-valued records in numeric feature columns were examined for semantic validity [8]. Features where zero is a meaningful value (e.g., failed login count) were retained unchanged; records with structurally invalid zero entries were imputed using column-wise median substitution to preserve distributional integrity.

Categorical Feature Encoding: Three features in NSL-KDD are categorical: protocol type (TCP, UDP, ICMP), service (http, ftp, smtp, etc., with 70 distinct values), and flag (11 distinct TCP connection states) [2], [9]. These were transformed using label encoding followed by one-hot encoding, converting each categorical variable into a set of binary indicator columns.

Feature Scaling: All continuous-valued features were normalized to the range [0, 1] using Min-Max scaling. Scaling parameters were fitted exclusively on the training set and applied to the test set, preventing data leakage [8].

Class Imbalance Mitigation: To address the severe underrepresentation of R2L and U2R categories, Synthetic Minority Oversampling Technique (SMOTE) is applied to the training set, generating synthetic interpolated samples

for minority classes until a configurable class ratio is achieved.

C. Hybrid Fusion Engine

The proposed IIDS employs a two-component Hybrid Fusion Engine that combines the outputs of a Shallow Learning Model (SLM) and a Deep Learning Model (DLM).

The SLM component consists of a Random Forest classifier trained on the full preprocessed 41-feature vector [1], [9]. Random Forest was selected for its robustness to irrelevant features, natural resistance to overfitting through ensemble bagging, and its ability to produce calibrated class probability estimates [9]. The model outputs a probability distribution PSLM ($y | x$) over the five class labels for each input record.

The DLM component is a CNN-LSTM architecture [5], [6], [7]. A 1D convolutional layer first extracts local spatial patterns from the feature vector, learning to detect co-occurring feature signatures characteristic of specific attack types [5]. The convolutional output is passed to a bidirectional LSTM layer, which models sequential dependencies across a sliding window of consecutive connection records, capturing temporally distributed attack patterns such as slow DoS or multi-step reconnaissance [5]. The Hybrid Fusion Engine combines both outputs through a learned weighted ensemble, where the fusion weight $\alpha \in [0,1]$ is optimized on a held-out validation set.

V. CONCLUSION

This paper proposes a hybrid Intelligent Intrusion Detection System (IIDS) that utilizes a combination of Random Forest and CNN-LSTM models to overcome some of the challenges associated with conventional intrusion detection systems [5], [6]. The proposed system, utilizing machine learning in conjunction with deep learning, is able to analyse and evaluate sophisticated network traffic and detect both known and unknown cyber threats. Using the NSL-KDD dataset [2], [8], the proposed model is able to achieve

superior detection accuracy, precision and recall.

The Random Forest part of the system is designed to effectively perform feature selection, thus reducing the dimensionality of the dataset and leading to improvement of overall computational efficiency [1], [9]. On the other hand, CNN-LSTM is designed to capture the spatial and temporal features of the data, allowing the system to detect attack patterns that typically might be overlooked by other systems [5], [7]. This hybrid design is able to improve the overall reliability while also reducing false alarms.

In this study, we demonstrated that combining a set of learning approaches makes intrusion detection systems more adaptive and resilient in the ever-changing cyber world. The value of these systems becomes more pronounced as threats grow more sophisticated [3], [4].

Future work would involve integrating advanced deep learning frameworks, processing real-time data streams applying the model to larger and more heterogeneous datasets and incorporating explainable AI methods to promote trust and reduce opacity in automated intrusion detection systems. In conclusion, the IIDS model is a valuable contribution to the development of intelligent and automated cybersecurity systems.

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